GVPT622 Lab Session

Statistical Inference

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University of Maryland

Set Up

- Load the packages we'll use today and set the seed:

```
library(readxl)
library(tidyverse)
library(janitor)
library(ggdist)
library(MetBrewer)
library(DescTools)

set.seed(1234)
```

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  - Sampling distributions and the Central Limit Theorem
  - Standard errors and confidence intervals
  - Choosing a sample size
  - A first hypothesis test

Population and Sample

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- **Parameter**: the number we care about in the population. We *don't* know it
- **Statistic**: the same number computed in our sample. We *do* know it
- Going from the statistic to the parameter is statistical inference
- From lecture: a poll of 800 likely Maine voters found 50% for Jackson. That 50% is a statistic; the true share among all likely Maine voters is the parameter

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- Large: the more observations, the closer we get to the truth (the law of large numbers)
  - More is always better for inference, if not for your budget
- Even a large, random sample comes with some error. Let's explore why

Sampling Error

Back to the Whip Count

- Last lab's undecided member votes yea or nay with equal probability. We watch 100 votes:

```
possible_outcomes <- c("YEA", "NAY")

votes_100 <- sample(possible_outcomes, 100, replace = T, prob = c(0.5,
    0.5))
table(votes_100)
```

```
votes_100
NAY YEA
 55  45
```

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```

- The true probability is 0.5, so we "should" see 50 yeas. We got 45. That gap is sampling error

Many 100-Vote Trials

- Let's run that 100-vote trial 10,000 times and count the yeas each time:

```
whip_count <- function(possible_outcomes, n) {

  outcomes <- sample(possible_outcomes, size = n, replace = T,
                     prob = c(0.5, 0.5))

  return(table(outcomes)["YEA"])

}

results <- tibble(trial = 1:1e4) |>
  rowwise() |>
  mutate(n_yea = whip_count(possible_outcomes, 100)) |>
  ungroup()
```

Many 100-Vote Trials

```
results
```

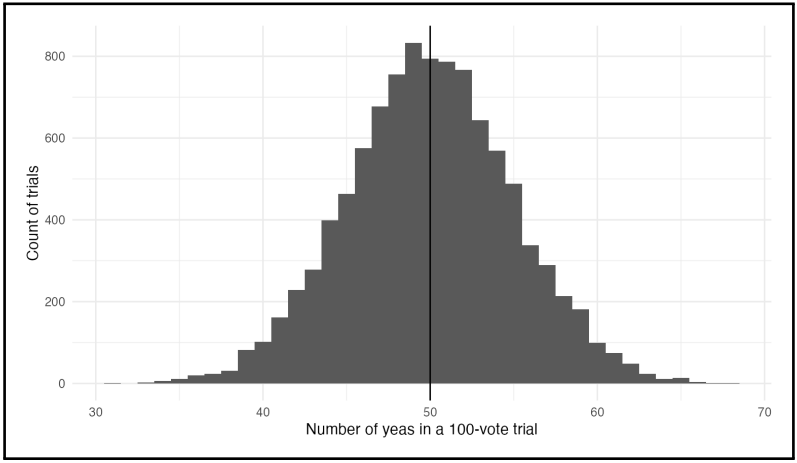
```
# A tibble: 10,000 x 2
   trial n_yea
   <int> <int>
 1     1    52
 2     2    52
 3     3    48
 4     4    60
 5     5    61
# i 9,995 more rows
```

(Output trimmed to 5 rows.) Each row is one 100-vote trial. Same member, same 50/50 probability, different answers.

The Distribution of Trials

```
ggplot(results, aes(x = n_yea)) +
  geom_histogram(binwidth = 1) +
  geom_vline(xintercept = 50) +
  theme_minimal() +
  labs(x = "Number of yeas in a 100-vote trial",
       y = "Count of trials")
```

The Distribution of Trials



Most trials land near 50 yeas, but some land far from it.

```
mean(results$n_yea)
range(results$n_yea)
```

```
[1] 49.9702
[1] 31 68
```

- Some trials were way off: as few as 31 yeas, as many as 68

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range(results$n_yea)
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- Some trials were way off: as few as 31 yeas, as many as 68
- But most cluster around 50, and the average is 49.97, almost exactly the truth

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```

- Some trials were way off: as few as 31 yeas, as many as 68
- But most cluster around 50, and the average is 49.97, almost exactly the truth
- That's what unbiased means: any one trial can miss, but the center of all the trials is the true value

Sampling Distributions

Case Study: Democrats in the House

- What share of House members are Democrats?

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- Then we'll pretend we don't, draw samples, and see how close the samples get

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- The CEL data cover every member in every Congress from the 93rd to the 118th (1973–2024): 11,606 member-terms
- So for once we actually have the population, and we can compute the true parameter
- Then we'll pretend we don't, draw samples, and see how close the samples get
- Note: a member who served five terms appears five times. Our population is member-terms, not people

Preparing the Data

```
cel <- read_excel("CELHouse93to118.xlsx", guess_max = 20000)

house <- cel |>
  rename(
    congress = "Congress number",
    name = "Legislator name, as given in THOMAS",
    dem = "1 = Democrat",
    female = "1 = female"
  ) |>
  select(congress, name, dem, female)
```

- dem and female are 0/1 indicators. We'll use female at the end of lab

The Population

```
head(house, 5)
```

```
# A tibble: 5 x 4
  congress name              dem female
     <dbl> <chr>           <dbl>  <dbl>
1       93 Abdnor, James       0      0
2       93 Abzug, Bella        1      1
3       93 Adams, Brock        1      0
4       93 Addabbo, Joseph     1      0
5       93 Albert, Carl        1      0
```

```
tabyl(house, dem)
```

```
 dem    n   percent
   0 5317 0.4581251
   1 6289 0.5418749
```

- 54.19% of all member-terms were Democrats. This is the parameter

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 dem    n   percent
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```

- 54.19% of all member-terms were Democrats. This is the parameter
- In research, you rarely get to see this

One Random Sample

- Draw a simple random sample of 1,000 member-terms. Every member-term is equally likely to be picked:

```
house_sample <- slice_sample(house, n = 1000)
tabyl(house_sample, dem)
```

```
 dem   n percent
   0 446   0.446
   1 554   0.554
```

One Random Sample

- Draw a simple random sample of 1,000 member-terms. Every member-term is equally likely to be picked:

```
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```

```
 dem   n percent
   0 446   0.446
   1 554   0.554
```

- 55.4%. Close to 54.19%, but not exactly

A Different Sample

```
house_sample_2 <- slice_sample(house, n = 1000)
tabyl(house_sample_2, dem)
```

```
 dem   n percent
   0 457   0.457
   1 543   0.543
```

- 54.3%. A different sample gives a different answer, just like our different 100-vote trials

A Different Sample

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```

- 54.3%. A different sample gives a different answer, just like our different 100-vote trials
- This is sampling variability: a statistic changes from sample to sample because of random sampling error

1,000 Samples of 1,000

```
dem_survey <- function(df, n) {

  slice_sample(df, n = n) |>
    tabyl(dem) |>
    filter(dem == 1) |>
    pull(percent)

}

house_samples_1000 <- tibble(survey = 1:1000) |>
  rowwise() |>
  mutate(prop_dem = dem_survey(house, 1000)) |>
  ungroup()
```

- dem_survey() draws one sample of size n and returns the share of Democrats in it

1,000 Samples of 1,000

```
house_samples_1000
```

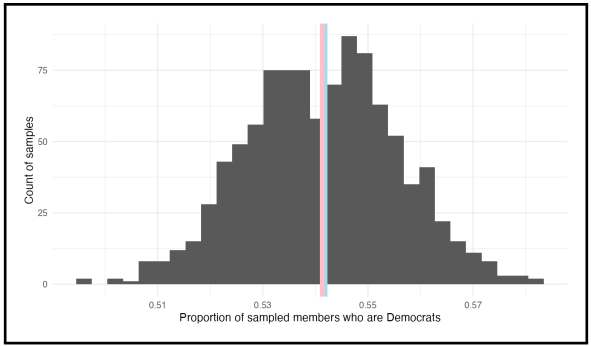
```
# A tibble: 1,000 x 2
   survey prop_dem
    <int>    <dbl>
 1      1    0.528
 2      2    0.541
 3      3    0.533
 4      4    0.549
 5      5    0.533
# i 995 more rows
```

```
ggplot(house_samples_1000, aes(x = prop_dem)) +
  geom_histogram() +
  geom_vline(xintercept = tabyl(house, dem) |> filter(dem == 1) |>
    pull(percent),
             colour = "lightblue",
             linewidth = 2) +
  geom_vline(xintercept = mean(house_samples_1000$prop_dem),
             colour = "pink",
             linewidth = 2) +
  theme_minimal() +
  labs(x = "Proportion of sampled members who are Democrats",
       y = "Count of samples")
```

The Sampling Distribution

```
mean(house_samples_1000$prop_dem)
```

```
[1] 0.541279
```



Pink: mean of the 1,000 samples (54.13%). Blue: the population parameter (54.19%).

- The average of our 1,000 sample proportions is 54.13%; the truth is 54.19%. With infinitely many samples, they'd match exactly

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- The average of our 1,000 sample proportions is 54.13%; the truth is 54.19%. With infinitely many samples, they'd match exactly
- The sampling distribution is the distribution of a statistic across all possible samples of the same size
- The Central Limit Theorem: for large, random samples, the sampling distribution is approximately Normal, centered on the population parameter
- This holds for means, proportions, and counts, even though each member's dem is just a 0 or a 1

Inferring From a Single Sample

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- In real research we rarely get 1,000 samples. You can't rerun the 118th Congress
- And we usually don't know the population's size or shape
- Most of the time we have one statistic: one mean, one proportion, one count
- What happens if we pick just one of our 1,000 samples?

Pick One Sample at Random

```
house_single <- slice_sample(house_samples_1000)
house_single
```

```
# A tibble: 1 x 2
  survey prop_dem
   <int>    <dbl>
1    628    0.523
```

- Sample 628 says 52.3%, about 2 points under the truth

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- Sample 628 says 52.3%, about 2 points under the truth
- Not perfect, but close. Since most samples sit near the center, a random pick is more likely to be close than far
- But we could get unlucky. How confident can we be in any one sample?

How Confident Can We Be?

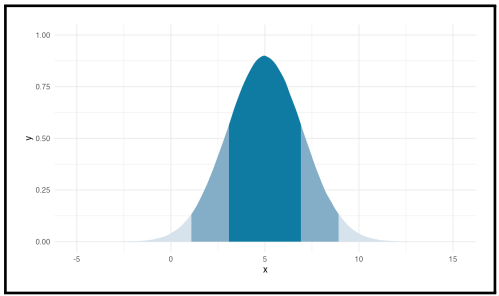
Properties of the Normal Distribution

```
norm_5_2 <- tibble(x = rnorm(n = 1e6, mean = 5, sd = 2))

ggplot(norm_5_2, aes(x = x)) +
  stat_slab(
    aes(fill_ramp = after_stat(cut_cdf_qi(cdf))), fill =
    met.brewer("Egypt")[2]
  ) +
  scale_fill_ramp_discrete(range = c(1, 0.2), guide = "none") +
  theme_minimal()
```

- A Normal distribution with mean 5 and standard deviation 2, drawn 1,000,000 times

Properties of the Normal Distribution



- About 68% of the data fall within 1 SD of the mean
- About 95% fall within 1.96 (≈ 2) SDs
- About 99.7% fall within 3 SDs

The bands cut_cdf_qi() draws are the middle 66% (dark), the middle 95% (medium), and everything else (light). Close to the 68/95 rule, but not identical.

From One Sample to a Sampling Distribution

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- The standard deviation of a sampling distribution is called the standard error. For a proportion:

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$$se(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- $\hat{p}$ ("p-hat") is the sample proportion. The hat marks a sample statistic

Our One Sample

```
my_sample <- slice_sample(house, n = 1000)

p_hat <- tabyl(my_sample, dem) |>
  filter(dem == 1) |>
  pull(percent)
p_hat

n <- 1000

se <- sqrt((p_hat * (1 - p_hat)) / n)
se
```

```
[1] 0.541
[1] 0.01575814
```

- A fresh sample: $\hat{p} = 54.1\%$, with a standard error of 1.6 points

68% and 95% Intervals

```
lower_68 <- p_hat - 1 * se
lower_68

upper_68 <- p_hat + 1 * se
upper_68
```

```
[1] 0.5252419
[1] 0.5567581
```

```
lower_95 <- p_hat - 1.96 * se
lower_95

upper_95 <- p_hat + 1.96 * se
upper_95
```

```
[1] 0.510114
[1] 0.571886
```

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upper_68 <- p_hat + 1 * se
upper_68
```

```
[1] 0.5252419
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lower_95 <- p_hat - 1.96 * se
lower_95

upper_95 <- p_hat + 1.96 * se
upper_95
```

```
[1] 0.510114
[1] 0.571886
```

- 95% interval: 51.0% to 57.2%. It contains the true 54.19%

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- What "95% confident" means: if we repeated this method on many samples, about 95% of the intervals would contain the true parameter
  - It's a statement about the method. Any one interval either contains the parameter, or it doesn't

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You Are the Expert

- All of this leans hard on one sample. That's fine if the sample is large and random
- If it isn't random, the sampling distribution isn't centered on the parameter. That's bias, and a bigger sample won't fix it
- Know your data and your subject. The better you know them, the sooner you'll notice when a sample or a finding looks off

Choosing Your Sample Size

Standard Error and Sample Size

- n is in the denominator of $se(\hat{p})$: bigger samples, smaller standard errors, narrower intervals. But by how much?

```
se_from_sample <- function(n_samples) {

  house |>
    slice_sample(n = n_samples) |>
    pull(dem) |>
    MeanSE(na.rm = T)

}
```

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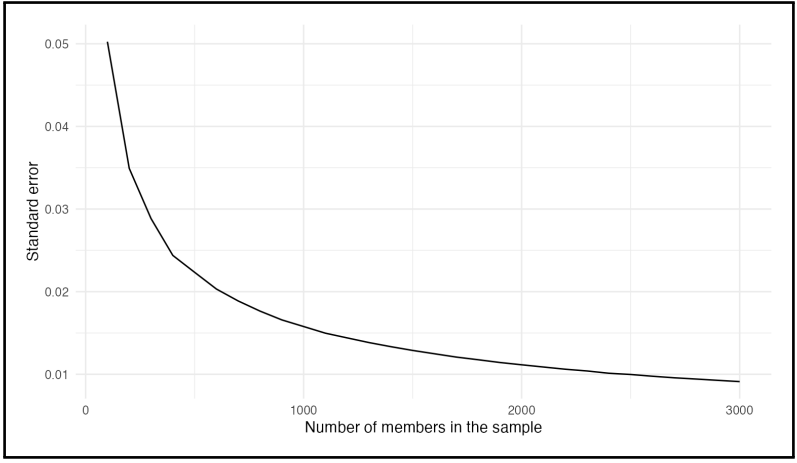
}
```

- MeanSE() (from DescTools) returns the standard error of a mean. The mean of a 0/1 variable is a proportion

Diminishing Returns

```
tibble(sample_size = seq(from = 100, to = 3000, by = 100)) |>
  rowwise() |>
  mutate(se = se_from_sample(sample_size)) |>
  ggplot(aes(x = sample_size, y = se)) +
  geom_line() +
  theme_minimal() +
  labs(x = "Number of members in the sample",
       y = "Standard error")
```

Diminishing Returns



Steep at first, then flat: each extra member buys a smaller drop in the standard error.

Diminishing Returns

- 100 $\rightarrow$ 1,000 members: the standard error drops from about 5 points to about 1.6

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- Why? se shrinks with $\sqrt{n}$, so to halve your margin of error you need four times the sample
  - From lecture: the Maine poll's ± 3.5 at $n = 800$ only halves to ± 1.7 at $n = 3{,}200$

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- 100 $\rightarrow$ 1,000 members: the standard error drops from about 5 points to about 1.6
- 1,000 $\rightarrow$ 1,900: it only drops to about 1.1
- Why? se shrinks with $\sqrt{n}$, so to halve your margin of error you need four times the sample
  - From lecture: the Maine poll's ± 3.5 at $n = 800$ only halves to ± 1.7 at $n = 3{,}200$
- More is always better, but every extra respondent buys you less, and survey costs keep adding up

Hypothesis Testing

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  - Does same-day registration increase turnout?

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- A hypothesis is a measurable, observable statement about a population
  - Does trade dependence make war between states less likely?
  - Does same-day registration increase turnout?
- The frequentist approach: imagine a world in which you're wrong. That's the null hypothesis (H_0)

Hypotheses

- A hypothesis is a measurable, observable statement about a population
  - Does trade dependence make war between states less likely?
  - Does same-day registration increase turnout?
- The frequentist approach: imagine a world in which you're wrong. That's the null hypothesis (H_0)
- Then ask: if H_0 were true, how likely would we be to see data like ours?

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- The catch: we only get our one sample of 1,000 member-terms, my_sample
- (Next week: comparing two groups, with difference-of-means and difference-of-proportions tests)

Women in Our Sample

```
my_women <- filter(my_sample, female == 1)
tabyl(my_women, dem)
```

```
 dem  n   percent
   0 37 0.2936508
   1 89 0.7063492
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- 126 women in our sample, and 70.6% of them are Democrats
- That looks like support for H_a. But only 126 people: could this just be random noise?

```
prop_women_dem <- mean(my_women$dem)
se_women <- sqrt((prop_women_dem * (1 - prop_women_dem)) /
    nrow(my_women))
se_women
```

```
[1] 0.04057327
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- With only 126 observations, the standard error is 4.1 points, much wider than the 1.6 we had with 1,000

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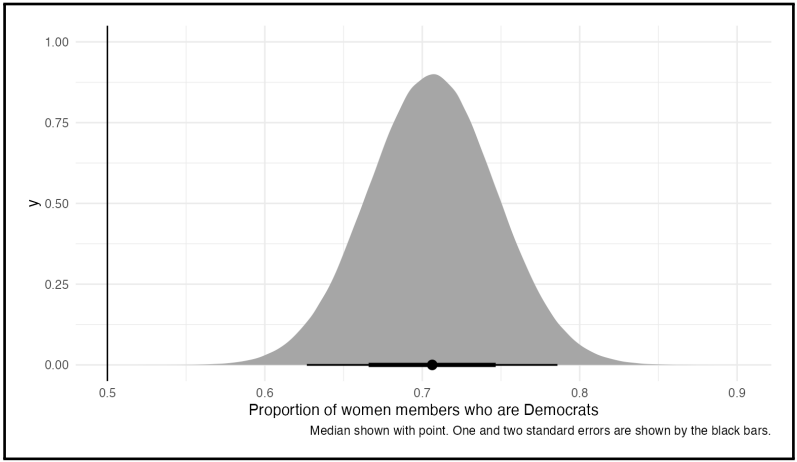
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- With only 126 observations, the standard error is 4.1 points, much wider than the 1.6 we had with 1,000
- 95% interval: $0.706 \pm 1.96 \times 0.041 = 62.7\%$ to 78.6%

Adding the Null World

```
ggplot(tibble(x = rnorm(1e6, mean = prop_women_dem, sd = se_women)),
       aes(x = x)) +
  stat_halfeye(.width = c(0.95, 0.68)) +
  geom_vline(xintercept = 0.5) +
  theme_minimal() +
  labs(x = "Proportion of women members who are Democrats",
       caption = "Median shown with point. One and two standard errors
    are shown by the black bars.")
```

Adding the Null World



The null value (0.5, vertical line) is far outside the 95% interval.

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- It sits far outside our 95% interval (62.7% to 78.6%). Almost none of our hypothetical samples come anywhere near it
- If H_0 were true, it would be very unlikely to draw a random sample in which 70.6% of women are Democrats
- How unlikely? We can put a number on it

Test Statistic and p-value

```
se_null <- sqrt((0.5 * (1 - 0.5)) / nrow(my_women))
z <- (prop_women_dem - 0.5) / se_null
z

2 * pnorm(-abs(z))
```

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[1] 4.632528
[1] 3.612271e-06
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- Test statistic z: how many standard errors our estimate sits from the null value. Here, 4.6. (We use the SE under the null, with $p = 0.5$)

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- Test statistic z: how many standard errors our estimate sits from the null value. Here, 4.6. (We use the SE under the null, with $p = 0.5$)
- p-value: the probability of a result at least this extreme if H_0 were true.

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- We reject the null: the share of women who are Democrats is statistically significantly different from 0.5
- This lines up with the interval: 0.5 is outside the 95% confidence interval, so we reject at the 0.05 level
- A small p-value says the data would be surprising under H_0. It doesn't tell us *why* women members skew Democratic

Checking Our Answer

- Unlike in real research, we can look at the whole population:

```
house |>
  filter(female == 1) |>
  tabyl(dem)
```

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 dem    n   percent
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- The true share is 71.5%. Our sample of 126 said 70.6%, and the 95% interval (62.7% to 78.6%) caught it

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Conclusion

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- The Central Limit Theorem: that sampling distribution is approximately Normal
- The standard error gives its spread. $\hat{p} \pm 1.96 \times se$ is a 95% confidence interval
- Standard errors shrink with $\sqrt{n}$: diminishing returns
- Hypothesis testing: how surprising would our data be in a world where we're wrong?

Citation

Center for Effective Lawmaking. 2025. "House Legislative Effectiveness Scores, 93rd–118th Congresses." `https://thelawmakers.org`.

Volden, Craig, and Alan E. Wiseman. 2014. *Legislative Effectiveness in the United States Congress: The Lawmakers*. New York: Cambridge University Press.

Questions?